

Bilkent University

Department of Electrical and Electronics Engineering

EE-102: Introduction to Digital Circuit Design

Lab 3 Report: Combinational Logic Circuit

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Date: February 19, 2026

1 Objectives

The purpose of this experiment was to design, build, and verify a logic circuit on a breadboard without the use of an FPGA. The specific objectives were:

- To understand the 74HC163 4-bit binary counter.
- To implement a Boolean function using discrete logic gates (74HC08 AND, 74HC32 OR).
- To simulate a truth table input stream using a counter driven by a signal generator.
- To verify the circuit functionality using an oscilloscope and a visual LED monitor system.

2 Design and Theory

2.1 Logic Function

Based on the design requirements, a function was chosen to be implemented using these two types of gates. (first two gates used on my previous vivado code). The function has 3 inputs which are the first 3 outputs of the counter (Q_0, Q_1, Q_2):

$$Y = (Q_0 \cdot Q_1) + Q_2 \quad (1)$$

Where:

- Q_0 (Pin 14) is the Least Significant Bit (LSB).
- Q_1 (Pin 13) is Bit 1.
- Q_2 (Pin 12) is Bit 2.

2.2 Truth Table (Expected)

The theoretical truth table for the function is derived below. Logic 1 represents High Voltage (5V). Logic 0 represents Low Voltage (0V).

Table 1: Theoretical Truth Table

Decimal	Q_2 (Input C)	Q_1 (Input B)	Q_0 (Input A)	Term 1 ($Q_0 \cdot Q_1$)	Output Y
0	0	0	0	0	0
1	0	0	1	0	0
2	0	1	0	0	0
3	0	1	1	1	1
4	1	0	0	0	1
5	1	0	1	0	1
6	1	1	0	0	1
7	1	1	1	1	1

Theoretical Duty Cycle: The output should be LOW for 3 counts and HIGH for 5 counts (3/8 Low, 5/8 High).

3 Materials and Equipment

- **ICs:** 74HC163 (4-bit Counter), 74HC08 (Quad 2-Input AND), 74HC32 (Quad 2-Input OR).
- **Equipment:** DC Power Supply (5V), Signal Generator, Oscilloscope.
- **Components:** Breadboards (x2), LEDs (Red), Resistors (560 Ω), Jumper Wires.

4 Procedure and Implementation

Phase 1: Counter Setup (74HC163)

- **Power:** Pin 16 was connected to +5V and Pin 8 to Ground.
- **Mode Configuration:** The chip was set to Count Mode by connecting the following control pins to High (+5V):
 - $\overline{\text{CLR}}$ (Pin 1) and $\overline{\text{LOAD}}$ (Pin 9) to disable resets.
 - ENP (Pin 7) and ENT (Pin 10) to enable counting.
- **Clock Input:** A 1 Hz square wave (0V to 5V) was generated and applied to Pin 2 (CLK).
- **Verification:** The output at Pin 14 (Q_0) was probed with an oscilloscope to confirm a 0.5 Hz square wave (half the clock frequency).

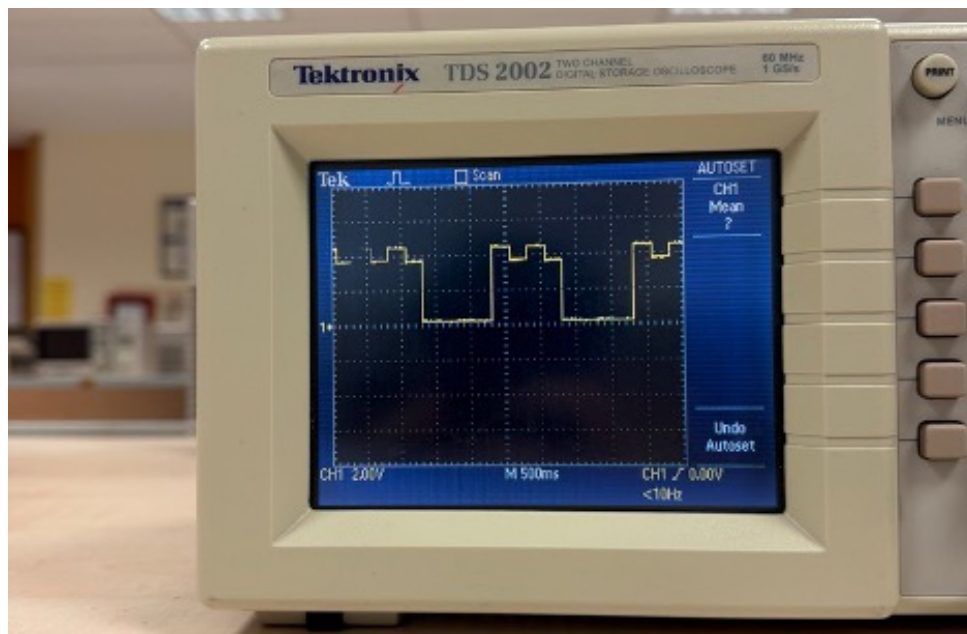
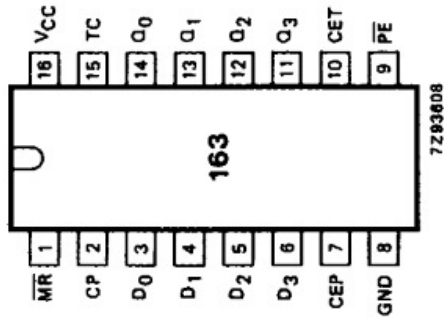


Figure 1: Oscilloscope Verification of Q_0 Output (0.5 Hz)



(a) Counter Overview

PIN NO.	SYMBOL	NAME AND FUNCTION
1	MR	synchronous master reset (active LOW)
2	CP	clock input (LOW-to-HIGH, edge-triggered)
3, 4, 5, 6	D ₀ to D ₃	data inputs
7	CEP	count enable input
8	GND	ground (0 V)
9	PE	parallel enable input (active LOW)
10	CET	count enable carry input
11, 12, 13, 14	Q ₀ to Q ₃	flip-flop outputs
15	TC	terminal count output
16	V _{CC}	positive supply voltage

(b) Counter Close-up

Figure 2: Counter Connection Diagrams

Phase 2: Logic Circuit Construction

- **Chip Placement:** The 74HC08 (AND) and 74HC32 (OR) chips were placed and powered (Pin 14 to +5V, Pin 7 to GND).
- **Wiring the Function:**
 - Counter Pin 14 (Q_0) and Pin 13 (Q_1) were connected to the inputs of the AND gate.
 - The output of the AND gate and Counter Pin 12 (Q_2) were connected to the inputs of the OR gate.
 - The final output was connected to a Red LED.

Phase 3: Monitor System

To visualize the inputs in real-time, a secondary "Monitor Board" was constructed:

- Three LEDs were wired in an Active High configuration (Signal → LED Anode → Resistor → Ground).
- The LEDs were connected to Q_2 , Q_1 , and Q_0 respectively to display the binary count (000 to 111).
- Power and Ground rails were bridged between the main board and the monitor board.

5 Troubleshooting and Debugging

Several issues were encountered and resolved during the lab:

Counter Not Counting: Initially, the oscilloscope showed a flat line.

- *Cause:* The control pins (Reset/Load) were connected to Ground, holding the chip in reset.
- *Solution:* Pins 1, 9, 7, and 10 were explicitly connected to the +5V rail.

Signal Generator Offset: The counter failed to respond to the clock.

- *Cause:* The signal generator was set to an AC signal (-2.5V to +2.5V), which is outside the TTL logic range.
- *Solution:* Offset was adjusted to +2.5VDC to create a 0V-to-5V logic signal.

Component Overheating: The AND gate chip became extremely hot.

- *Cause:* The AND gate chip was connected reversed
- *Solution:* The orientation of the AND gate was fixed immediately.

LED Not Lighting: - *Cause:* The LED was plugged in backward, and the resistor was connected to the wrong power line.

- *Solution:* We flipped the LED around and moved the resistor to the negative (ground) rail. This allowed the electricity to flow properly from the chip's output, through the LED, and into the ground, lighting it up correctly.

6 Results and Verification

6.1 Oscilloscope Verification

The output waveform at the final OR gate was captured.

Observation: The waveform showed a repeating pattern of Logic LOW for approximately 3 divisions and Logic HIGH for approximately 5 divisions.

Analysis: This matches the theoretical expectation of the function $Y = (Q_0 \cdot Q_1) + Q_2$, which outputs 0 for counts 0-2 and 1 for counts 3-7.



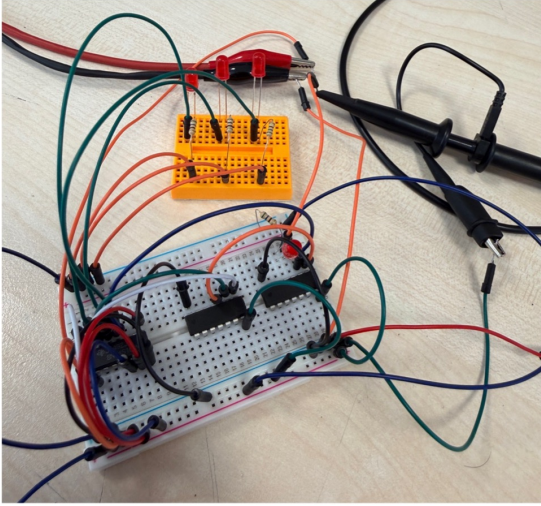
Figure 3: Oscilloscope Verification of Output Y

6.2 Visual Verification (1 Hz Test)

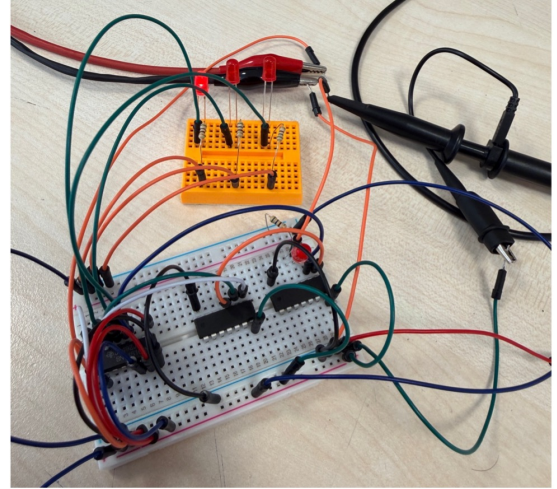
The clock frequency was lowered to 1 Hz to allow visual confirmation. The Monitor Board LEDs (Red) displayed the binary count while the Main Output LED (Red) displayed the result Y.

Table 2: Visual Verification Results

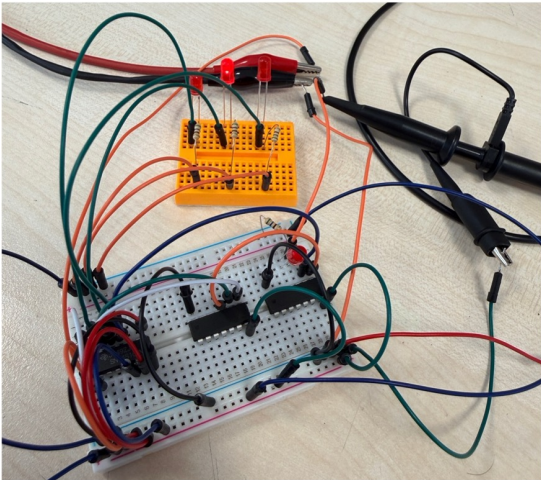
Count	Monitor Q_2	Monitor Q_1	Monitor Q_0	Output LED Y	Status
0	OFF	OFF	OFF	OFF	Verified
1	OFF	OFF	ON	OFF	Verified
2	OFF	ON	OFF	OFF	Verified
3	OFF	ON	ON	ON	Verified
4	ON	OFF	OFF	ON	Verified
5	ON	OFF	ON	ON	Verified
6	ON	ON	OFF	ON	Verified
7	ON	ON	ON	ON	Verified



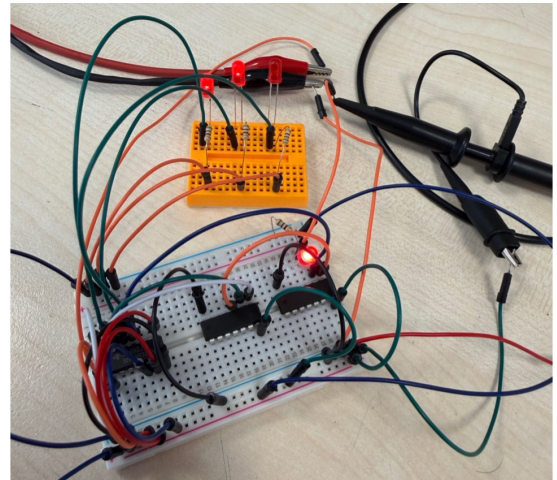
(a) Count 0



(b) Count 1

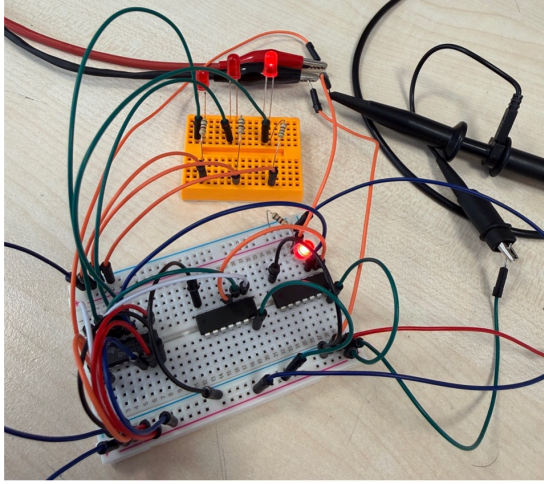


(c) Count 2

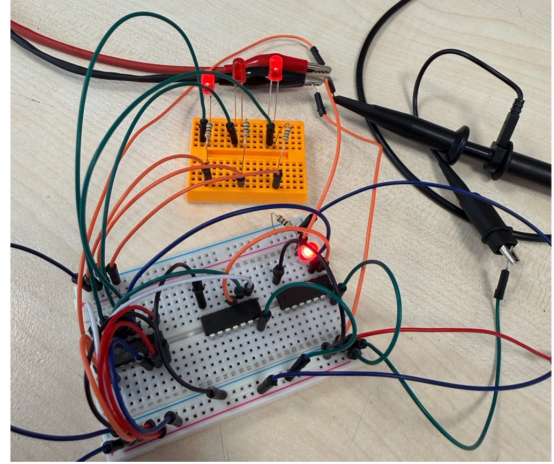


(d) Count 3

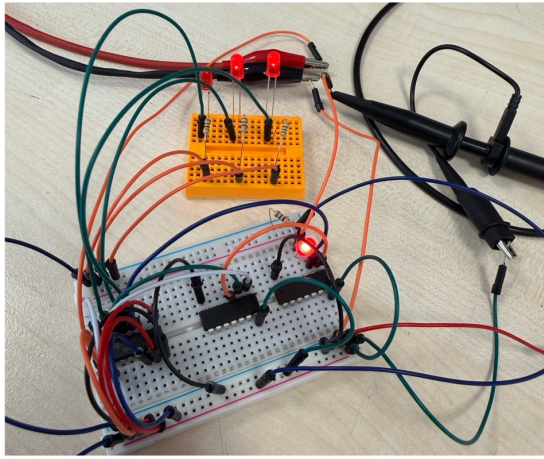
Figure 4: Visual Verification of States 0 through 3



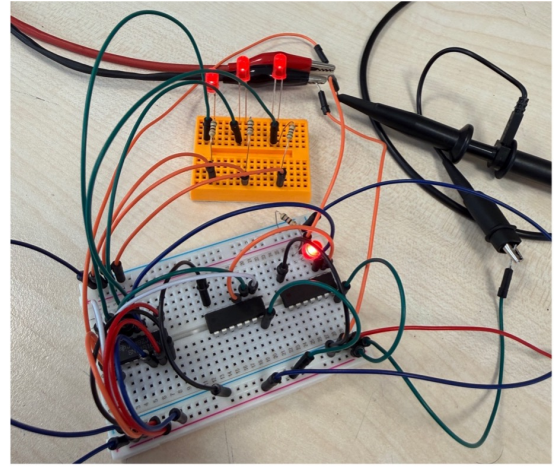
(a) Count 4



(b) Count 5



(c) Count 6



(d) Count 7

Figure 5: Visual Verification of States 4 through 7

7 Conclusion

The combinational logic circuit was successfully designed, built, and verified. The 74HC163 counter provided a reliable test vector source, eliminating the need for manual switching. The oscilloscope data confirmed the timing characteristics of the logic, and the visual monitor board confirmed the truth table state-by-state. All debugging issues, including power rail shorts and signal offsets, were resolved, resulting in a fully functional system.